An effective-ODE (averaging-theory) drift bound for one round of FederatedAveraging

Mini-study extending McMahan et al. (2017), arXiv:1602.05629

Abstract

For one round of FederatedAveraging (FedAvg) we idealize each client's u_k local full-batch gradient steps as the gradient flow of its local objective run for continuous time $T_k = \eta\, u_k$. We prove that the server map—the weight-averaged composition of these per-client flows—agrees with the gradient flow of the *global* objective f to second order in the time horizon, with the leading discrepancy equal to a curvature–gradient covariance across clients:

$$\Big\| \sum_k \beta_k \Phi_k^T(w) - \Phi_f^T(w) \Big\| = \tfrac{T^2}{2}\, \sigma_{HG} + O(T^3), \qquad \sigma_{HG} = \|\mathrm{Cov}_k(H_k, g_k)\|,$$

with an explicit cubic remainder ($C_3 = \rho G^2 + L^2 G$) under a bounded-third-derivative assumption. The leading $\frac{T^2}{2}$ term is independent of how the horizon T is split into local steps, so the per-round drift depends on (E, B, η) only through the single invariant $T = \eta E n_k / B$ (the discrete FedAvg map matches the flow up to an $O(T/u_k)$ Euler error, Lemma 2). The heuristic $\Delta = \eta^2\, \mathrm{Cov}_k(H_k, g_k)$ of the parent study's deep dive shares the *same* drift object $\Gamma = \mathrm{Cov}_k(H_k, g_k)$; at $E = 2$, $B = \infty$ ($T = 2\eta$) the flow gives $2\eta^2\Gamma$, twice the discrete two-step heuristic's $\eta^2\Gamma$—the factor 2 being exactly the Euler error of the 2-step sketch—and the correct prefactor $T^2/2 = (\eta u_k)^2/2$ grows as $u_k^2/2$ relative to the frozen η^2.

1 Setup

There are K clients with local objectives $F_k : \mathbb{R}^d \to \mathbb{R}$, weights $\beta_k \geq 0$, $\sum_k \beta_k = 1$, and global objective

$$f = \sum_{k=1}^{K} \beta_k F_k.$$

We write the gradient and Hessian at the broadcast point w as $g_k = \nabla F_k(w)$, $H_k = \nabla^2 F_k(w)$, $g = \nabla f(w) = \sum_k \beta_k g_k$, $H = \nabla^2 f(w) = \sum_k \beta_k H_k$. (We treat $w \in \mathbb{R}^d$; the matrix-parameter softmax model of the prototype is the vectorized case $w = \mathrm{vec}(W)$, $d = \dim W$, under the Frobenius inner product, to which everything applies verbatim.) Throughout, $\|\cdot\|$ is the Euclidean norm on vectors and the induced (operator) norm on the matrix $\nabla^2 F_k$ and the 3-tensor $\nabla^3 F_k$; these are compatible, so $\|Ax\| \leq \|A\|\,\|x\|$. For the matrix-valued H_k and vector-valued g_k we write $\mathrm{Cov}_k(H_k, g_k) := \mathbb{E}_k[H_k g_k] - \mathbb{E}_k[H_k]\,\mathbb{E}_k[g_k] \in \mathbb{R}^d$, the covariance taken *through* the matrix–vector product, with $\mathbb{E}_k$ the expectation under the weights β_k.

Assumption 1 (Smoothness)**.** There is a closed ball $\mathcal{B} \subset \mathbb{R}^d$ containing w and a constant $\rho < \infty$ such that each $F_k \in C^3(\mathcal{B})$ with $\sup_{v \in \mathcal{B}} \|\nabla^3 F_k(v)\| \leq \rho$ and $\sup_{v \in \mathcal{B}} \|\nabla F_k(v)\| \leq G$, $\sup_{v \in \mathcal{B}} \|\nabla^2 F_k(v)\| \leq L$.

We further assume w lies in the *interior* of $\mathcal{B}$. Since each flow has speed $\left\|\dot{\phi}_k^s\right\| = \|\nabla F_k\| \le G$ (and likewise $\left\|\dot{\psi}^s\right\| \le G$), there is a positive escape time $T_0 \ge \mathrm{dist}(w, \partial\mathcal{B})/G$ such that, for every $0 \le T \le T_0$, all client trajectories ϕ_k^s *and* the global trajectory ψ^s remain in $\mathcal{B}$ for $s \in [0, T]$. Define the gradient-flow maps

$$\Phi_k^T(w) = \phi_k^T, \quad \dot{\phi}_k^s = -\nabla F_k(\phi_k^s),\ \phi_k^0 = w; \qquad \Phi_f^T(w) = \psi^T, \quad \dot{\psi}^s = -\nabla f(\psi^s),\ \psi^0 = w. \tag{1}$$

The FedAvg *server map* is the weighted average of per-client flow maps,

$$\mathcal{S}^T(w) := \sum_{k=1}^{K} \beta_k\, \Phi_k^T(w). \tag{2}$$

The horizon $T = \eta u_k$ is the continuous-time image of $u_k = En_k/B$ discrete local steps of size η; see Lemma 2 for the discrete link.

2 Lie–Taylor expansion of a gradient flow

Lemma 1 (Third-order flow expansion)**.** *Under Assumption 1, for $0 \le T \le T_0$ (with T_0 chosen so the trajectory stays in $\mathcal{B}$),*

$$\Phi_k^T(w) = w - T\, g_k + \frac{T^2}{2}\, H_k g_k + R_k(T), \qquad \|R_k(T)\| \le \frac{T^3}{6}\, C_3, \tag{3}$$

where $C_3 := \rho G^2 + L^2 G$ bounds the third s-derivative of ϕ_k^s on $\mathcal{B}$. The same holds for Φ_f^T with (g_k, H_k) replaced by (g, H) and the same constant (since f is a convex combination of the F_k).

Proof. Let $V = -\nabla F_k$, so $\dot{\phi}^s = V(\phi^s)$. Then $\frac{d}{ds}\phi^s = V(\phi^s)$ and, differentiating again, $\frac{d^2}{ds^2}\phi^s = DV(\phi^s)\, V(\phi^s)$, where $DV = -\nabla^2 F_k$. At $s = 0$: $\phi^0 = w$, $\dot{\phi}^0 = -g_k$, $\ddot{\phi}^0 = (-H_k)(-g_k) = H_k g_k$. Taylor's theorem with the Lagrange/integral remainder gives (3) with $R_k(T) = \frac{1}{2}\int_0^T (T-s)^2\, \dddot{\phi}^{\,s}\, ds$ in the integral form, so $\|R_k(T)\| \le \frac{T^3}{6} \sup_{s \le T} \left\|\dddot{\phi}^{\,s}\right\|$. Differentiating a third time,

$$\dddot{\phi}^{\,s} = \frac{d}{ds}\big[DV(\phi^s)V(\phi^s)\big] = D^2V(\phi^s)\big(V(\phi^s), V(\phi^s)\big) + DV(\phi^s)DV(\phi^s)V(\phi^s),$$

and $\left\|D^2V\right\| = \left\|\nabla^3 F_k\right\| \le \rho$, $\|DV\| = \left\|\nabla^2 F_k\right\| \le L$, $\|V\| = \|\nabla F_k\| \le G$ on $\mathcal{B}$, so $\left\|\dddot{\phi}^{\,s}\right\| \le \rho G^2 + L^2 G = C_3$. The statement for Φ_f^T is identical because $\nabla^3 f = \sum_k \beta_k \nabla^3 F_k$ obeys the same bound by the triangle inequality, using $\beta_k \ge 0$ and $\sum_k \beta_k = 1$ (no convexity of the objectives is assumed or used anywhere). $\square$

3 Main result

Theorem 1 (Effective-ODE drift bound)**.** *Let Assumption 1 hold and suppose all clients share the same horizon $T_k \equiv T$ (equal effective time; e.g. equal client sizes). Then for $0 \le T \le T_0$,*

$$\mathcal{S}^T(w) - \Phi_f^T(w) = \frac{T^2}{2}\, \Gamma + R(T), \qquad \Gamma := \sum_k \beta_k H_k g_k - Hg = \mathrm{Cov}_k(H_k, g_k), \tag{4}$$

with $\|R(T)\| \le \frac{T^3}{3} C_3$*. Consequently, writing* $\sigma_{HG} := \|\Gamma\|$*,*

$$\Big|\ \big\|\mathcal{S}^T(w) - \Phi_f^T(w)\big\| - \frac{T^2}{2}\,\sigma_{HG}\ \Big| \ \le\ \frac{T^3}{3}\,C_3, \qquad i.e. \qquad \big\|\mathcal{S}^T(w) - \Phi_f^T(w)\big\| = \frac{T^2}{2}\,\sigma_{HG} + O(T^3), \tag{5}$$

and the leading-order drift direction is exactly $\Gamma/\|\Gamma\|$*.*

Proof. Average (3) over k with weights β_k:

$$\mathcal{S}^T(w) = \sum_k \beta_k \Phi_k^T(w) = w - T\sum_k \beta_k g_k + \frac{T^2}{2}\sum_k \beta_k H_k g_k + \sum_k \beta_k R_k(T) = w - Tg + \frac{T^2}{2}\sum_k \beta_k H_k g_k + R_1(T),$$

with $\|R_1(T)\| = \|\sum_k \beta_k R_k(T)\| \le \sum_k \beta_k \frac{T^3}{6} C_3 = \frac{T^3}{6} C_3$ (Jensen / triangle inequality, $\sum_k \beta_k = 1$). By Lemma 1 applied to f, $\Phi_f^T(w) = w - Tg + \frac{T^2}{2} Hg + R_2(T)$ with $\|R_2(T)\| \le \frac{T^3}{6} C_3$. Subtracting, the $O(1)$ term w and the $O(T)$ term $-Tg$ cancel exactly, leaving

$$\mathcal{S}^T(w) - \Phi_f^T(w) = \frac{T^2}{2}\Big(\sum_k \beta_k H_k g_k - Hg\Big) + \big(R_1(T) - R_2(T)\big) = \frac{T^2}{2}\,\Gamma + R(T),$$

$\|R(T)\| \le \|R_1\| + \|R_2\| \le \frac{T^3}{3} C_3$. The covariance identity is the algebra

$$\Gamma = \sum_k \beta_k H_k g_k - \Big(\sum_k \beta_k H_k\Big)\Big(\sum_k \beta_k g_k\Big) = \mathbb{E}_k[H_k g_k] - \mathbb{E}_k[H_k]\,\mathbb{E}_k[g_k] = \mathrm{Cov}_k(H_k, g_k),$$

where $\mathbb{E}_k$ is expectation under the probability weights β_k. Finally (5) is the reverse triangle inequality applied to (4). □

Remark 1 (Unequal horizons)**.** If client horizons differ, $T_k = \eta u_k$ with $u_k = En_k/B$, assume $\max_k T_k \le T_0$ (so every $\phi_k^{T_k}$ and the reference $\psi^{\bar{T}}$ stay in $\mathcal{B}$), and define the weighted mean horizon $\bar{T} = \sum_k \beta_k T_k$ and the heterogeneous server map $\mathcal{S}_{\text{het}}(w) := \sum_k \beta_k \Phi_k^{T_k}(w)$. Then $\mathcal{S}_{\text{het}}(w) - \Phi_f^{\bar{T}}(w) = -\sum_k \beta_k (T_k - \bar{T}) g_k + \frac{1}{2}\big(\sum_k \beta_k T_k^2 H_k g_k - \bar{T}^2 Hg\big) + O(T^3)$. The new first-order term $-\sum_k \beta_k (T_k - \bar{T}) g_k = -\mathrm{Cov}_k(T_k, g_k)$ is the *size-heterogeneity* drift; it vanishes when client sizes are equal ($T_k \equiv T$, recovering Theorem 1) or when one compares against the size-weighted reference $\sum_k \beta_k \Phi_f^{T_k}$. Either way the result is governed entirely by the horizons $\{T_k = \eta E n_k/B\}$, never by (E, B, η) separately.

Lemma 2 (Discrete-to-flow consistency)**.** *Let* $\widehat{\Phi}_k^T$ *be the* u_k*-step explicit Euler discretization of* (1) *with step* $\eta = T/u_k$ *(i.e. literal full-batch local GD). Then* $\Big\|\widehat{\Phi}_k^T(w) - \Phi_k^T(w)\Big\| \le \frac{\eta}{2} G\,(e^{LT} - 1) = \frac{GT(e^{LT}-1)}{2u_k} = O(\eta) \to 0$ *as* $u_k \to \infty$ *at fixed* T*. Hence the discrete FedAvg server map satisfies the bound* (5) *up to an additional* $O(\eta) = O(T/u_k)$ *term, which the prototype confirms vanishes as* E *grows at fixed* T*.*

Proof. Standard global-error bound for explicit Euler on a field with Jacobian bound L and second-derivative-of-solution bound $\big\|\ddot{\phi}\big\| \le LG$ (since $\ddot{\phi} = DV\,V$ with $\|DV\| \le L$, $\|V\| \le G$): the global error is $\le \frac{\eta}{2L}(e^{LT} - 1)\sup\big\|\ddot{\phi}\big\| \le \frac{\eta}{2L}(e^{LT} - 1)\,LG = \frac{\eta}{2} G(e^{LT} - 1)$, which is $O(\eta) = O(T/u_k)$ for LT bounded. Averaging over k with $\sum_k \beta_k = 1$ preserves the bound. □

4 Consequences

Effective outer ODE. Iterating Theorem 1, the server sequence $w_{t+1} = \Phi_f^T(w_t) - \frac{T^2}{2}\Gamma(w_t) + O(T^3)$ is, in the slow time $s = tT$, a consistent one-step discretization of the modified (backward-error) flow below: over a slow-time interval $[0, S]$ the $N = S/T$ steps accumulate $N \cdot O(T^3) = O(T^2)$ error, so to first order in T the iterates track the perturbed flow

$$\frac{dw}{ds} = -\nabla f(w) - \frac{T}{2}\,\mathrm{Cov}_k\left(H_k, g_k\right) + O(T^2).$$

The heterogeneity field $\mathrm{Cov}_k(H_k, g_k)$ is the continuous-time avatar of the deep-dive's Δ; it vanishes when clients are statistically identical ($H_k \equiv H$ or $g_k \equiv g$), recovering exact centralized gradient flow.

Recovering and correcting the heuristic. For $E = 2$, $B = \infty$ we have $u_k = 2$, $T = 2\eta$, and Theorem 1 gives drift $\frac{(2\eta)^2}{2}\sigma_{HG} = 2\eta^2\sigma_{HG}$ with the *same* object $\Gamma = \mathrm{Cov}_k(H_k, g_k)$ as the two-step Taylor sketch. This flow value $2\eta^2$ is exactly twice the discrete two-step heuristic η^2; the factor 2 is the explicit-Euler global error of Lemma 2 at $u_k = 2$, and it shrinks as $O(1/u_k)$ once E grows. For general E the correct prefactor is $T^2/2 = (\eta u_k)^2/2$, not the frozen η^2: relative to the raw η^2 heuristic the magnitude grows by $u_k^2/2$ (a factor 2 already at $E = 2$ and 12.5 at $E = 5$). The single invariant is $T = \eta E n_k/B$.

A Softmax cross-entropy satisfies Assumption 1

For mean cross-entropy with softmax probabilities $p_i = \mathrm{softmax}(x_i^\top W)$, the gradient is $\frac{1}{m}\sum_i x_i(p_i - y_i)^\top$ and the Hessian acts as $H[V] = \frac{1}{m}\sum_i x_i\big((\mathrm{diag}(p_i) - p_i p_i^\top)(x_i^\top V)\big)^\top$ (used verbatim in the prototype's exact Hessian-vector product). All derivatives are bounded on *all* of W-space because $p \mapsto$ softmax is C^∞ with bounded derivatives ($\|p\| \le 1$) and the data $\{x_i\}$ are fixed; concretely $G \le \sqrt{2}\,\max_i \|x_i\|$, $L \le \max_i \|x_i\|^2$, and $\rho \le 2\max_i \|x_i\|^3$. Since these hold on every ball, one may take $\mathcal{B}$ large enough around w that the speed-$\le G$ flow stays inside for $T \le T_0 = \mathrm{dist}(w, \partial\mathcal{B})/G > 0$; hence Assumption 1 holds in full (both the constants and the trajectory-confinement).